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Central bank reputation with noise

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ABSTRACT

This paper presents a model of central bank reputation where inflation is a noisy function of central bank actions. In the model, a central bank can be "hawkish", with a relatively high penalty for taking inflationary actions, or "dovish", with a relatively low penalty. The central bank's type varies according to a Markov process, and its reputation is the Bayesian posterior of households that it is a hawkish type. We show analytically that no Markov pooling equilibria exist: hawkish and dovish central banks must act differently. We then argue that without sufficient noise, it is difficult for Markov separating equilibria to exist either. But *with* sufficient noise, we show computationally that pure strategy separating equilibria exist and have appealing characteristics: both hawkish and dovish banks choose lower inflationary actions than they would in the absence of reputation considerations, and both types are most aggressive in attempting to gain a hawkish reputation when their current reputation is middling.

1. Introduction

This paper presents a simple model of central bank reputation with several appealing characteristics not common in reputation games. In it, a benevolent central bank can be a "hawkish" type, which receives a relatively large negative payoff to taking inflationary actions, or a "dovish" type, which receives a lesser negative payoff to taking inflationary actions. Notably, neither type is behavioral, and each can continuously choose an action that affects the distribution of inflation outcomes. This implies that unlike games where the hawkish type is behavioral, our game lets us consider how a hawkish central bank "should" act to recover a hawkish reputation if it has lost one, or maintain a hawkish reputation if it has one.

The basic model builds on the work of Chari et al. (1998), where sellers in a monopolistic competition environment have market power when setting posted prices, and are embedded in a cash-in-advance economy where fiat money creation is a noisy function of central bank actions. The fundamental tension in such an economy is that because of market power and the cash-in-advance constraint, the real prices of consumption goods, relative to labor, are set inefficiently high, which creates a temptation for the central bank to use inflation to reduce these inefficiencies ex-post. (In equilibrium, however, such money-printing-induced inflation is anticipated by price setters and only exacerbates the inefficiencies in the game.) Our contribution is to add uncertainty by households over the type of central bank they are facing. This allows for reputation to be defined as the Bayesian posterior of households that they are facing a hawkish central bank.

We show first, analytically, that no Markov pooling equilibria in our game exist. We then argue that without sufficient noise disconnecting a central bank's target inflation from realized inflation, it is difficult for a pure symmetric Markov separating equilibria to exist. Nevertheless, we then show computationally that with sufficient noise, a pure strategy symmetric Markov

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separating equilibrium exists and has appealing characteristics. In particular, in equilibrium, both types of central banks choose lower inflationary actions than they would in the absence of reputational considerations, and both central bank types are most aggressive in attempting to gain a hawkish reputation when their current reputation is middling.

The intuition behind the no pooling result is simple: if both types of central banks act the same, realized inflation isn't informative and thus doesn't affect tomorrow's reputation, and thus there are no dynamic incentives to inflating more or less. But by assumption, hawkish and dovish central banks have different static incentives, and thus an equilibrium can't have them acting the same.

The intuition behind why noise is useful for existence is more subtle: If a hawkish central bank inflates less than a dovish central bank, a high reputation is valuable to both types, since inefficient household markups are increasing in expected inflation. This implies the dovish central bank is tempted to deviate and mimic the hawkish type and gain its reputation. Without sufficient noise, doing so is relatively cheap — one period of the dovish type taking the hawkish action moves reputation considerably. In fact, with no noise and separating strategies, one period of the dovish type taking the hawkish action moves reputation to its extreme. Using noise to partially disconnect central bank actions from resulting inflation outcomes makes it more costly for a dovish central bank to mimic a hawkish central bank since it must do so repeatedly.

That both types inflate less in equilibrium than they would absent reputation considerations happens for the simple reason that reputation is valuable. Both hawkish and dovish central banks understand that high inflation hurts their reputation (and thus continuation payoff). That this effect is largest when reputation is middling (as opposed to households being almost convinced the central bank is one or the other type) comes directly from Bayes' rule — it is difficult to move an extreme reputation and easier to move an interior reputation. Knowing this, in equilibrium, both central bank types act accordingly.

2. Related literature

There is a large literature on reputation and monetary policy. Building on models of central bank behavior that emphasize the time inconsistency of the optimal policy (Kydland and Prescott, 1977; Calvo, 1978; Barro and Gordon, 1983; Rogoff, 1985), this paper starts with the monopolistic competition framework of Chari et al. (1998). Similarly to many earlier contributions (Backus and Driffill, 1985; Barro, 1986; Vickers, 1986), we introduce uncertainty among households regarding the type of central bank they face, allowing reputation to be defined as the Bayesian posterior of households identifying a "hawkish" central bank type. We do so in an infinite horizon model where the types of the central bank are forever changing, as in Ball (1995) and Phelan (2006). We allow for imperfect monitoring of the central bank actions by introducing noise between the central bank monetary choice and its corresponding equilibrium outcome (Canzoneri, 1985; Cukierman and Meltzer, 1986; Cukierman, 2000; Faust and Svensson, 2001; Sleet, 2001; Laubach, 2003). We do not study announcements, but our setup is related to papers that analyze the role of monetary announcements with imperfect information about the central bank type (Cukierman and Liviatan, 1991; Schultz, 1996; Walsh, 2000). Work by King et al. (2008) extends these analyses by focusing on a model with a "strong" central bank type that can commit to its announcements and a "weak" one that can deviate. Lu et al. (2016) follows up with a New Keynesian model with imperfect monitoring where the "weak" central bank type follows a pre-specified behavioral rule. A more recent paper by Kostadinov and Roldán (2025) studies the role of announcements in a New Keynesian model where the "weak" type optimizes. Dovis and Kirpalani (2021) introduce a third agent, the rule designer, responsible for recommending the inflation target. Other recent works on monetary policy commitment are Afrouzi et al. (2023) and Schreger et al. (2024).

3. Environment

In this section, we present a version of the model in Chari et al. (1998), modified to allow for noisy signals and two types of central banks (and thus a role for reputation).

Time is discrete and infinite with $t \in \{0, 1, \dots\}$. The real economy consists of a mass 1 continuum of households, with names $j \in [0, 1]$, split into shopper–seller pairs. In each period, each household j produces good j using only its individual effort. Households consume a composite good made up of the output off all households so that the consumption of the composite good by household j, c_j, is such that

$$c_j = \left(\int_k c_{j,k}^{\frac{\sigma-1}{\sigma}} \, dk \right)^{\frac{\sigma}{\sigma-1}},$$

where $c_{j,k}$ is the consumption of household j of good k, and where $\sigma > 1$. A household j that consumes c_j units of the composite good and produces y_j of its own good receives a static payoff $\log(c_j) - \alpha y_j$. In a symmetric *efficient* allocation, all households consume the same amount of each good at each date, $c^* = \frac{1}{\alpha}$.

Our solution concept will be pure symmetric Markov perfect equilibrium of a game between households and central banks.

3.1. Players

The players of our game consist of long-lived households and central banks of varying types.

As stated previously, we assume a continuum of households of mass 1, with names $j \in [0, 1]$, split into shopper–seller pairs, where each household j produces good j.

A central bank can be one of two types, $i \in \{1, 2\}$. Type 1 is "hawkish", or receives a relatively large penalty for taking inflationary actions, while type 2 is "dovish", receiving a lower such penalty. (Both types care about the welfare of households as well.)

Initially, with probability $\rho_0 \in [0,1]$, the central bank is of type 1. After $t = 0$, the type of central bank follows a one stage Markov process. If the current central bank is of type 1, it is replaced next period by a central bank of type 2 with probability $\delta \in [0,1]$. Likewise, a central bank of type 2 is replaced next period by a central bank of type 1 with probability $\epsilon \in [0,1]$. At all times, central bank type is private.[1]

Importantly, we assume that when a central bank leaves the game, it never comes back. That is, if a type 1 bank leaves the game, replaced by a type 2 bank, and then later that type 2 bank is replaced by a type 1 bank, that new type 1 bank is a *different player* than the previous type 1 bank. When we later define payoffs, central banks will care only about outcomes while they are active in the game. This substantially simplifies our analysis.

3.2. Timing

At the beginning of period t, a household j starts off with initial holdings of money, $M_{j,t}$, which it brings from the previous period. Let M_t be the mean value, over households, of $M_{j,t}$. The household then chooses the nominal price of its product, $P_{j,t}$, which obligates the seller to produce whatever quantity other households choose to purchase at that price.

After households individually set their prices, the central bank (of type i) privately chooses an action $\mu_{i,t} \in [0,\overline{\mu}]$ (where $\overline{\mu}$ may be infinity) which probabilistically affects the rate of money growth.

After the central bank chooses $\mu_{i,t}$, the actual rate of money growth $\mu_{a,t}$ is realized according to a distribution $F(\mu_{a,t}|\mu_{i,t})$, with support $[0,\infty)$ for all μ and with density $f(\mu_a|\mu)$ differentiable in μ. Upon the realization of $\mu_{a,t}$, household j's shopper is given $M_t\mu_{a,t}$ units of newly created money lump sum.[2]

Next, shopper j with $M_{j,t}+M_t\mu_{a,t}$ units of money chooses how much to purchase of each good k, $c_{j,k,t}$, subject to a cash-in-advance constraint that

$$\int_k P_{k,t} c_{j,k,t} dk \le M_{j,t} + M_t \mu_{a,t}.$$

The money holdings of household j at the end of the period are then whatever was not spent by the household's shopper plus whatever cash was collected by the household's seller in return for producing good j.

Finally, at the end of the period, type switches occur as described in the previous section.

3.3. Pure symmetric Markov strategies

We will consider only **pure** symmetric Markov strategies. Strategies are symmetric if, for any history of the game, all households set the same price, consume the same amount of each good, and consume the same amount as every other household. Since every household is assumed to start the game at $t = 0$ with the same amount of money, or $M_{j,0} = M_0$, in such a symmetric specification, after every history, every household will hold the same amount of money as every other household, or $M_{j,t} = M_t$ for all j. Our equilibrium conditions will ensure that such symmetry is individually rational. By Markov, we mean that households and central banks choose their actions as a function of history only to the extent that the history affects the probability that households assign to the current central bank being of type 1, denoted by ρ. In particular, it means that the stock of money M_t won't matter once prices are scaled relative to it.

To this end, let $P(\rho)$ represent the common price set by all households (as a function of ρ), relative to common household money holdings M_t. That is, if at date t all households hold M_t units of money, all households set the price of their good to $M_t P(\rho)$.

Let $\mu_1(\rho)$ and $\mu_2(\rho)$ represent the actions of each central bank type, again as a function of ρ.

Finally, let $c(\mu_a,\rho)$ represent household consumption (same across all goods) as a function of *beginning of period beliefs*, ρ, and the realized rate of money growth, μ_a. Since households choose consumption after μ_a is realized, the realization of μ_a, along with beginning period beliefs ρ, determine household beliefs regarding bank type when their consumption decision is made. Nevertheless, since $c(\mu_a,\rho)$ is also a function of μ_a directly, it is without loss to make household consumption a function of beginning-of-period beliefs ρ, as opposed to updated middle-of-period beliefs. Likewise, since prices are quoted relative to the beginning-of-period stock of money, M, and the increase in the money supply, μ_a, is also relative to M, the absolute stock of money, M, does not matter in determining consumption.

Summing up, a pure symmetric Markov strategy is a specification

$$(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho)).$$

[1] It is not central to our analysis that δ is possibly different than ϵ. In our benchmark computed example in Section 8, they are set equal to each other. The main benefit to allowing them to differ is it allows us to vary one but not the other when checking robustness, as we do in Section 8.

[2] The assumption that $\mu_{a,t}$ must be non-negative is made solely to avoid the possibility that off-path, a household does not have enough money to pay a lump sum tax.

3.4. Beliefs

Our assumption that the support of μ_a is the same for all μ_i ensures that Bayes' rule can always be used to determine beliefs. To this end, let $\hat{\rho}(\mu_a,\rho)$ be the Bayesian posterior after μ_a is observed but before end-of-period type switches, and $\rho^+(\mu_a,\rho)$ be the Bayesian posterior at the beginning of the next period, after a possible type switch realization. Then,

$$\hat{\rho}(\mu_a,\rho)=\frac{\rho f(\mu_a|\mu_1(\rho))}{\rho f(\mu_a|\mu_1(\rho))+(1-\rho)f(\mu_a|\mu_2(\rho))},$$

and

$$\rho^+(\mu_a,\rho)=(1-\delta)\hat{\rho}(\mu_a,\rho)+\epsilon(1-\hat{\rho}(\mu_a,\rho)). \tag{1}$$

3.5. Payoffs

Household j, which produces y_j and consumes $c_{j,k}$ for goods $k\in[0,1]$, gets a utility flow given by

$$\log(c_j)-\alpha y_j,$$

where

$$c_j=\left(\int_k c_{j,k}^{\frac{\sigma-1}{\sigma}}\,dk\right)^{\frac{\sigma}{\sigma-1}}.$$

Over time and uncertainty, a household cares about the expectation of its discounted, by $\beta\in(0,1)$, lifetime per-period utility stream.

If all households set the same price and consume the same amount of each good, then $y_j=y$ for all j, and $c_{j,k}=c=y$ for all (j,k). Given this, we assume a central bank of type i that chooses action μ_i gets static payoff

$$\log(c)-\alpha c-\gamma_i h(\mu_i).$$

Here, $h(\mu_i)$ represents a negative payoff to a central bank for taking an inflationary action against, say, a mandate not to do so. (We assume $h(\mu_i)$ is twice differentiable with $h(0)=0$, $h_i'(0)=0$, $h'(\mu_i)>0$ and $h''(\mu_i)>0$, for $\mu_i>0$.) Central bank types i differ by γ_i, the amount of disutility they get by choosing $\mu_i>0$. We assume $\gamma_1>\gamma_2$, which means that a type 1 central bank is the "hawkish" central bank.

Over time and uncertainty, central banks care about the expectation of their discounted per-period stream of payoffs while they are active in the game. For type 1 banks, we define their discount factor, β_1, to be $\beta_1=\beta(1-\delta)$, and for type 2 banks, we define their effective discount factor to be $\beta_2=\beta(1-\epsilon)$.[3]

3.6. Equilibrium conditions

The household problem. Consider the problem of a household that starts the period with M_j units of money when all other agents start with M, and the aggregate state is ρ. Given an aggregate state ρ, we let $c(\mu_a,\rho)$ and $P(\rho)$ denote the aggregate responses of consumption and prices, and let $\rho^+(\mu_a,\rho)$ denote its evolution.

In a symmetric equilibrium all other sellers set the same price for their goods, and thus it is optimal for this household's shopper to spend the same amount on each good as a function of the realized rate of money growth μ_a, $\hat{c}(\mu_a)$. The household must also decide how much money to carry to the next period as a function of the realized rate of money growth, $M^+(\mu_a)$, how much to produce, $\hat{y}(\mu_a)$, and the price of its good, $\hat{P}$. The measurability constraint on $\hat{P}$ (that it cannot be a function of μ_a) reflects the fact that the seller sets the price before the realization of μ_a. Abusing notation and letting $F(\mu_a|\rho)=\rho F(\mu_a|\mu_1(\rho))+(1-\rho)F(\mu_a|\mu_2(\rho))$, we can write the problem of a household recursively as follows:

$$V(M_j|M,\rho)=\max_{\{\hat{c}(\mu_a),M^+(\mu_a),\hat{y}(\mu_a),\hat{P}\}}\int\left[\log(\hat{c}(\mu_a))-\alpha\hat{y}(\mu_a)+\beta V\left(M^+(\mu_a)|M(1+\mu_a),\rho^+(\mu_a,\rho)\right)\right]dF(\mu_a|\rho),$$

subject to:

$$MP(\rho)\hat{c}(\mu_a)\le M_j+M\mu_a,$$

$$M^+(\mu_a)=M_j+M\mu_a-MP(\rho)\hat{c}(\mu_a)+M\hat{P}\hat{y}(\mu_a),$$

$$\hat{y}(\mu_a)=c(\mu_a,\rho)\left(\frac{\hat{P}}{P(\rho)}\right)^{-\sigma},$$

[3] We have assumed that a central bank type is replaced by an opposite type, but it is possible to relax this. Suppose that χ_1 and χ_2 are the probabilities that the central bank is replaced by its own type. The effective discount factors of the central banks would be $\beta_1=\beta(1-\delta-\chi_1)$, and $\beta_2=\beta(1-\epsilon-\chi_2)$. As long as these switches remain private information, nothing else is affected: the households problem remains just as in what follows, and Bayes' rule is unchanged (as the presence of χ_i does not affect the updating rule).

and $\hat{P} \geq 0, \hat{c}(\mu_a) \geq 0, M^+(\mu_a) \geq 0$.

The first constraint in this problem is the cash-in-advance constraint of the shopper. The second constraint is the budget constraint of the household which shows that future money holdings equal the money holdings inherited from the previous period plus the central bank's transfers and minus the difference between expenditure and income. The third constraint is the restriction that the seller must satisfy the demand at the quoted price, $\hat{P}$.

Let $\hat{c}(\mu_a, M_j, M, \rho)$, $\hat{P}(M_j, M, \rho)$, $M^+(\mu_a, M_j, M, \rho)$, and $\hat{y}(\mu_a, M_j, M, \rho)$ denote the associated optimal policy functions. Then, in an equilibrium, these must coincide with the aggregate behavior when $M_j = M$; that is, $\hat{c}(\mu_a, M, M, \rho) = c(\mu_a, \rho)$, and $\hat{P}(M, M, \rho) = P(\rho)$. Note that as a result, $M^+(\mu_a, M, M, \rho) = (1 + \mu_a)M$ and $\hat{y}(\mu_a, M, M, \rho) = c(\mu_a, \rho)$.

Define the expected marginal utility of consumption, $g(\rho)$, to be

$$g(\rho) \equiv \int \frac{1}{c(\mu_a, \rho)} dF(\mu_a|\rho).$$

Using the envelope condition of the household problem, we get that

$$V'(M|M, \rho) = \frac{g(\rho)}{MP(\rho)}.$$

Using the first order condition with respect to $\hat{P}$, and imposing $\hat{P} = P(\rho)$, and $\hat{c}(\mu_a) = c(\mu_a, \rho)$, we obtain that

$$\int \left\{ \alpha\sigma \frac{c(\mu_a, \rho)}{P(\rho)} + \beta(1 - \sigma)c(\mu_a, \rho)MV'(M(1 + \mu_a)|M(1 + \mu_a), \rho^+(\mu_a, \rho)) \right\} dF(\mu_a|\rho) = 0.$$

Then, using the envelope condition, $V'(M(1 + \mu_a)|M(1 + \mu_a), \rho^+(\mu_a, \rho)) = \frac{g(\rho^+(\mu_a,\rho))}{M(1+\mu_a)P(\rho^+(\mu_a,\rho))}$, we obtain that

$$\int \left\{ \frac{\alpha\sigma}{P(\rho)} + \frac{\beta(1 - \sigma)}{1 + \mu_a} \frac{g(\rho^+(\mu_a, \rho))}{P(\rho^+(\mu_a, \rho))} \right\} c(\mu_a, \rho) dF(\mu_a|\rho) = 0. \tag{2}$$

Eq. (2) thus ensures that $P(\rho)$ is optimally chosen by households. We next consider ensuring $c(\mu_a, \rho)$ is optimally chosen by households.

If $M_j = M$ and the cash-in-advance constraint does not bind, $c(\mu_a, \rho) < \frac{1+\mu_a}{P(\rho)}$; thus, for $c(\mu_a, \rho)$ to satisfy household optimization, it must satisfy the first order condition:

$$\frac{1}{c(\mu_a, \rho)} = \frac{\beta}{1 + \mu_a} \frac{P(\rho)}{P(\rho^+(\mu_a, \rho))} g(\rho^+(\mu_a, \rho)),$$

in which case

$$c(\mu_a, \rho) = \frac{1 + \mu_a}{P(\rho)} \left(\frac{P(\rho^+(\mu_a, \rho))}{\beta} \frac{1}{g(\rho^+(\mu_a, \rho))} \right),$$

which implies that

$$\frac{P(\rho^+(\mu_a, \rho))}{\beta} \frac{1}{g(\rho^+(\mu_a, \rho))} < 1,$$

since $c(\mu_a, \rho) < (1 + \mu_a)/P(\rho)$.

If the cash-in-advance constraint binds, $c(\mu_a, \rho) = \frac{1+\mu_a}{P(\rho)}$, then optimality requires that

$$\frac{1}{c(\mu_a, \rho)} \geq \frac{\beta}{1 + \mu_a} \frac{P(\rho)}{P(\rho^+(\mu_a, \rho))} g(\rho^+(\mu_a, \rho)),$$

in which case

$$c(\mu_a, \rho) \leq \frac{1 + \mu_a}{P(\rho)} \left(\frac{P(\rho^+(\mu_a, \rho))}{\beta} \frac{1}{g(\rho^+(\mu_a, \rho))} \right),$$

which implies

$$\frac{P(\rho^+(\mu_a, \rho))}{\beta} \frac{1}{g(\rho^+(\mu_a, \rho))} \geq 1.$$

Thus, the optimality condition for household consumption can be summarized by

$$c(\mu_a, \rho) = \frac{1 + \mu_a}{P(\rho)} \min \left\{ \frac{P(\rho^+(\mu_a, \rho))}{\beta} \frac{1}{g(\rho^+(\mu_a, \rho))}, 1 \right\}, \tag{3}$$

reflecting whether the cash-in-advance constraint binds or not.

Eqs. (2) and (3) summarize the optimality conditions of the household problem. Note the current stock of money, M, is absent in both conditions, due to prices being defined as relative to M. That is, the actual price set by households is $MP(\rho)$ not $P(\rho)$.

Central bank's problem. Consider a central bank of type $i \in \{1, 2\}$ that must choose μ_i when its reputation is ρ. Let $V_i(\rho)$ be the lifetime payoff to a central bank of type i. Recall $\beta_1 = \beta(1-\delta)$ and $\beta_2 = \beta(1-\epsilon)$ are the effective discount rates for each central bank type. Given this, the decision problem for the type i bank when choosing μ_i is[4]

$$V_i(\rho) = \max_{\mu_i \in [0,\overline{\mu}]} \int \left((1-\beta_i)(\log(c(\mu_a, \rho)) - \alpha c(\mu_a, \rho) - \gamma_i h(\mu_i)) + \beta_i V_i(\rho^+(\mu_a, \rho)) \right) dF(\mu_a | \mu_i). \tag{4}$$

Thus, the local optimality condition for $\mu_i(\rho)$, if we assume that $\mu_i(\rho)$ is interior, is[5]

$$\begin{aligned} &(1-\beta_i)\frac{\partial \mathbb{E}[\log(c(\mu_a,\rho)) - \alpha c(\mu_a,\rho) | \mu = \mu_i(\rho)]}{\partial \mu} \\ &+\beta_i \frac{\partial \mathbb{E}[V_i(\rho^+(\mu_a,\rho)) | \mu = \mu_i(\rho)]}{\partial \mu} = (1-\beta_i)\gamma_i h'(\mu_i(\rho)). \end{aligned} \tag{5}$$

3.7. Equilibrium definition

We are now ready to define an equilibrium.

Definition 1. A pure symmetric Markov equilibrium is a strategy $(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho))$, and a belief update function, $\rho^+(\mu_a, \rho)$, where $\rho \in [\epsilon, 1-\delta]$ and $\mu_a \in [0, \infty)$, such that (i) Bayes' rule holds, that is, Eq. (1) holds given $\mu_1(\rho)$, and $\mu_2(\rho)$; (ii) the household optimality conditions (2) and (3) hold; (iii) $\mu_i(\rho)$ solves Problem (4) for $i \in \{1, 2\}$.

With an equilibrium thus defined, inflation is determined as follows: Suppose at a given date t the money supply is M_t, the central bank's reputation is ρ, and realized money growth is μ_a. The price level at date t is then $M_t P(\rho)$, while the price level at date $t+1$ is $M_t(1+\mu_a)P(\rho^+(\mu_a, \rho))$, so inflation between dates t and $t+1$ is

$$(1+\mu_a)\frac{P(\rho^+(\mu_a, \rho))}{P(\rho)} - 1, \tag{6}$$

which is notably not a function of M_t. In words, if reputation stays constant, or $\rho^+(\mu_a, \rho) = \rho$, inflation is simply the growth rate in the money supply, μ_a. If reputation changes due to the realization of the growth rate of money, or $\rho^+(\mu_a, \rho) \neq \rho$, inflation is also affected by the effect of this changed reputation on seller prices relative to the money supply, $P(\rho)$.

4. A single central bank type

In this section, we consider the case where there is only one type of central bank to make clearer the basic forces of the model when reputation is not an issue, both with and without noise.

4.1. The case without noise

In the simplest special case, the central bank's type is known and permanently fixed, or $\rho_0 = 1$ and $\epsilon = \delta = 0$, and there is no noise, or $\mu_a = \mu$ with certainty. (The latter violates our assumption that the density $f(\mu_a|\mu)$ exists, but we can nevertheless analyze this case.)

Here, a pure symmetric Markov strategy is simply a specification of two scalars and function, $(P, \mu, c(\mu_a))$ with $c(\mu_a) > 0$ and $P > 0$. The scalar P represents the constant price, *relative to the beginning period money supply*, μ represents the constant action of the central bank — in this case, the percent increase in the money supply — and $c(\mu_a)$ represents the consumption of households not only for the specified increase in the money supply, μ, but also for all possible values, μ_a. Even though $\mu_a = \mu$ with certainty, the specification of the consumption response of households to off-path increases in the money supply is necessary to check the incentives of the central bank.

In this simplified case, the optimality condition for household consumption, from Eq. (3), is given by:

$$c(\mu_a) = \frac{1+\mu_a}{P} \min\left\{ \frac{Pc(\mu)}{\beta}, 1 \right\},$$

where we used that $g = \frac{1}{c(\mu)}$. Here, $c(\mu_a)$ on the left hand side is the specification of the optimal consumption of the household for arbitrary μ_a, whereas $c(\mu)$ on the right hand side is the specification of consumption in the subsequent period for the specified central bank action μ.

This implies

$$\frac{Pc(\mu_a)}{1+\mu_a} = \min\left\{ \frac{Pc(\mu)}{\beta}, 1 \right\}.$$

[4] Recall that we are assuming that central banks only care about outcomes while they are active. This simplifies the calculation of their payoff as otherwise we would have to define two value functions for each type, one for when they are active and a second for when the other type is active.

[5] In what follows, for any function ϕ, we define $\frac{\partial \mathbb{E}[\phi(\mu_a)|\mu=x]}{\partial \mu} \equiv \frac{\partial}{\partial \mu} \left[\int \phi(\mu_a) dF(\mu_a|\mu) \right] \Big|_{\mu=x}$.

Note that the left hand side is the fraction of cash spent for arbitrary μ_a and the right hand side is independent of μ_a, and thus household optimality implies the cash-in-advance constraint binds for either all μ_a or no μ_a. It follows that the cash-in-advance constraint must bind for all μ_a.[6] Then,

$$c(\mu_a) = \frac{1+\mu_a}{P}.$$

The equilibrium condition (2), the determination of the price, can be written as

$$\frac{\alpha\sigma}{P} + \frac{\beta(1-\sigma)}{1+\mu}\frac{g}{P} = 0.$$

Using again that $g = \frac{1}{c(\mu)}$, we obtain that

$$c(\mu) = \underbrace{\frac{1}{\alpha}}_{\text{efficiency}} \times \underbrace{\frac{\sigma-1}{\sigma}}_{\text{monopoly distortion}} \times \underbrace{\frac{\beta}{1+\mu}}_{\text{cash-in-advance distortion}} \tag{7}$$

Using now that $c(\mu) = \frac{1+\mu}{P}$, we see that the equilibrium P must be such that

$$P = (1+\mu)^2 \frac{\sigma}{\sigma-1}\frac{\alpha}{\beta}. \tag{8}$$

Eq. (7) implies that equilibrium prices are set so that consumption and production are inefficiently low (since $\frac{\sigma-1}{\sigma}\frac{\beta}{1+\mu} < 1$ and efficiency requires $c = \frac{1}{\alpha}$). This condition demonstrates two inefficiencies. First, since $\sigma \in (1,\infty)$, goods are less than perfect substitutes for each other, and consequently each household symmetrically exploits its monopoly power when setting prices. The second inefficiency is the standard cash-in-advance inefficiency. The term $\frac{\beta}{1+\mu} < 1$ reflects the effect on price setting today of the fact that the dollars received in a transaction today cannot be spent until tomorrow, when the price level is $(1+\mu)$ times higher than today.

For the equilibrium determination of μ, note first that without noise, the equilibrium condition (5) becomes

$$(1-\beta)\frac{\partial[\log(c(\mu)) - \alpha c(\mu)]}{\partial\mu} = (1-\beta)\gamma h'(\mu),$$

which simplifies to

$$\frac{1}{P}\left(\frac{1}{c(\mu)} - \alpha\right) = \gamma h'(\mu). \tag{9}$$

Hence, with an arbitrary P fixed, as the central bank increases μ, the marginal benefit is that it increases both consumption and production by $\frac{1}{P}$, which marginally affects its payoff by $\frac{1}{c(\mu)} - \alpha$, which it trades off with the penalty associated with choosing a higher μ. But note that from (7), the left hand side of (9) is strictly positive, and thus given $h'(0) = 0$, $\mu > 0$ in any pure symmetric Markov perfect equilibrium. That is, a money growth rate $\mu > 0$ and a constant, relative to the money supply, P, implies a positive constant inflation rate. The central bank's ability to inflate makes an inefficient equilibrium even worse: the ability of a central bank to print money increases the wedge between the marginal utility of consumption and the marginal disutility to production present in (7).

4.2. The case with noise

Next, allow noise, but still require the central bank's type to be known and permanently fixed. In this case, a pure symmetric Markov strategy is again a specification, $(P, \mu, c(\mu_a))$. Here, it is still the case that the cash-in-advance constraint binds for all μ_a realizations. To see this, note from (3) that

$$\frac{Pc(\mu_a)}{1+\mu_a} = \min\left\{\frac{P}{\beta g}, 1\right\}, \tag{10}$$

where $g = \mathbb{E}[\frac{1}{c(\mu_a)}|\mu]$. Here, the left hand side is the fraction of money holdings spent on current consumption, while the right hand side is a constant. Thus, as with the case with no noise (and a constant central bank type), either the cash-in-advance constraint binds for all μ_a or it binds for no μ_a. An argument similar to the above shows that the cash-in-advance constraint must bind for any realization of μ_a.[7]

[6] If the cash-in-advance constraint were not to bind, then we would have that $c(\mu) = \frac{1+\mu}{P}\frac{P}{\beta}c(\mu) = \frac{1+\mu}{\beta}c(\mu)$, which is impossible given $c(\mu) > 0$ and $(1+\mu)/\beta > 1$.

[7] Suppose the cash-in-advance constraint does not bind for a given $\mu_a \geq 0$. Then, Eq. (10) implies

$$\frac{1}{1+\mu_a} = \frac{1}{c(\mu_a)}\frac{1}{\beta g} \Rightarrow \mathbb{E}\left[\frac{1}{1+\mu_a}|\mu\right] = \mathbb{E}\left[\frac{1}{c(\mu_a)}|\mu\right]\frac{1}{\beta g} \Rightarrow \mathbb{E}\left[\frac{1}{1+\mu_a}|\mu\right] = g\frac{1}{\beta g} \Rightarrow \mathbb{E}\left[\frac{\beta}{1+\mu_a}|\mu\right] = 1,$$

a contradiction.

Note that simplifying the first order condition of the household regarding setting the price of its good, (2), to allow for only one type of central bank delivers

$$P = \alpha \frac{\sigma}{\sigma - 1} \frac{1}{\beta} \frac{\mathbb{E}[1 + \mu_a | \mu]}{\mathbb{E}[\frac{1}{1+\mu_a} | \mu]}. \tag{11}$$

Hence we can write equilibrium consumption, as a function the realized money growth rate μ_a, as

$$c(\mu_a) = \underbrace{\frac{1}{\alpha}}_{\text{efficiency}} \times \underbrace{\frac{\sigma - 1}{\sigma}}_{\text{monopoly distortion}} \times \underbrace{\beta \mathbb{E}\left[\frac{1}{1+\mu_a} | \mu\right]}_{\text{cash-in-advance distortion}} \times \underbrace{\frac{1 + \mu_a}{\mathbb{E}[1 + \mu_a | \mu]}}_{\text{randomness}} \tag{12}$$

It is useful to compare consumption as a function of inflation with and without noise. In both cases, efficiency dictates that consumption should equal $1/\alpha$, the first term on the right hand side in both Eqs. (7) and (12). Without noise, consumption deviates from efficiency for two reasons: the monopoly distortion and the cash-in-advance distortion, the second and third terms on the right hand sides of Eqs. (7) and (12). With noise a third deviation from efficiency is present: the random realization of μ_a also matters. This effect is captured by the last term on the right hand side of (12). Note, however, that if μ_a is realized to be its expectation or lower, consumption will be inefficiently low since $\sigma > 1$, $\beta < 1$, and $\mu_a \geq 0$. That consumption is, in expectation, inefficiently low creates an incentive for the central bank to inflate since realized consumption increases with inflation.

To guarantee that with noise a central bank sets $\mu > 0$ requires additional structure regarding how the central bank's action μ affects realized inflation μ_a relative to the severity of the pricing distortion. To this end, note that without types, under the assumption of Markov strategies, the continuation value of the central bank is itself a constant. Using $c(\mu_a) = (1 + \mu_a)/P$ together with the price in (11), we can write the (interior) equilibrium condition for central bank optimization (5) as

$$\frac{\partial \mathbb{E}[\log(1 + \mu_a) | \mu]}{\partial \mu} - \frac{\sigma - 1}{\sigma} \beta \frac{\mathbb{E}[\frac{1}{1+\mu_a} | \mu]}{\mathbb{E}[1 + \mu_a | \mu]} \frac{\partial \mathbb{E}[1 + \mu_a | \mu]}{\partial \mu} = \gamma h'(\mu).$$

The following assumption mechanically guarantees that the equilibrium μ is strictly positive.

Assumption 1. The density $f(\mu_a|\mu)$, the discount parameter β, and the elasticity of substitution parameter σ are such that

$$\frac{\partial \mathbb{E}[\log(1 + \mu_a) | \mu = 0]}{\partial \mu} > \left(\frac{\sigma - 1}{\sigma} \beta \frac{\mathbb{E}\left(\frac{1}{1+\mu_a} | \mu = 0\right)}{\mathbb{E}(1 + \mu_a | \mu = 0)} \right) \frac{\partial \mathbb{E}[1 + \mu_a | \mu = 0]}{\partial \mu}. \tag{13}$$

To understand this assumption, note from (11) that the expression

$$\frac{\sigma - 1}{\sigma} \beta \frac{\mathbb{E}[\frac{1}{1+\mu_a} | \mu = 0]}{\mathbb{E}[1 + \mu_a | \mu = 0]} \tag{14}$$

captures the (inverse) distortion in pricing generated by both the monopoly distortion and the cash-in-advance constraint when $\mu = 0$. That is, the lower σ, the lower β, and the higher expected inflation when $\mu = 0$, the higher pricing distortion, and the lower expression (14). Thus if one considers the natural case where a higher μ shifts the distribution of μ_a such that $\partial \mathbb{E}[(1 + \mu_a)|\mu = 0]/\partial \mu > 0$ and $\partial \mathbb{E}[\log(1+\mu_a)|\mu = 0]/\partial \mu > 0$, or higher μ generates both higher expected inflation and higher expected log inflation, then Assumption 1 becomes a requirement that the pricing distortion is sufficiently severe that the central banker's marginal incentive to increase μ from $\mu = 0$ is positive. This guarantees in any Markov equilibrium with noise, as is the case without noise, $\mu > 0$: after households set prices, the central bank wishes to inflate to mitigate the pricing distortion caused by monopoly power and the cash-in-advance constraint.[8]

5. Reputation about what?

In our general model, we have allowed the two types of central banks to differ in two ways. First, they have a different inflation costs, with type 1 being more hawkish than type 2: $\gamma_1 > \gamma_2$. But there is another difference: the central bank types may have different effective discount factors. This occurs when $\epsilon \neq \delta$ and thus $\beta_1 \neq \beta_2$.

Here, we show if $\gamma_1 = \gamma_2$, or the two central banks are equally hawkish, then even if they have different discount rates, a Markov equilibrium exists where reputation plays no role.

Proposition 1. *Suppose that $\gamma_1 = \gamma_2 = \gamma$. Let $(P^*, \mu^*, c^*(\mu_a))$ be such that (i) P^* is the value of P that satisfies Eq. (11) with $\mu = \mu^*$, (ii) $c^*(\mu_a)$ is the function $c(\mu_a)$ that satisfies Eq. (12) with $\mu = \mu^*$, and (iii) μ^* solves :*

$$\mu^* \in \operatorname*{argmax}_{\mu \in [0, \overline{\mu}]} \int \left\{ \log(c^*(\mu_a)) - \alpha c^*(\mu_a) - \gamma h(\mu) \right\} dF(\mu_a|\mu).$$

Then, $P(\rho) = P^*$, $\mu_1(\rho) = \mu_2(\rho) = \mu^*$ and $c(\mu_a, \rho) = c^*(\mu_a)$ is a Markov equilibrium.

[8] Note if $\mu_a = \mu$ with certainty (no noise), Assumption 1 is satisfied. The left hand side of (13) is one, while the right hand side is $\frac{\sigma-1}{\sigma}\beta < 1$.

Proof. Given that reputation ρ is conjectured not affect any of the equilibrium objects, the equilibrium value functions $V_i(\rho^+(\mu_a,\rho))$ must also be independent of ρ.

It follows then that the optimal action by any central bank must satisfy the static optimality condition stated in part (iii) of the proposition. And given that $\gamma_i = \gamma$, it is a best response for both types to choose the same action, μ^*.

But then, from the perspective of the households, the central bank always chooses the same action. Thus if (11) and (12) hold given the central bank action μ^*, households are optimizing given the arguments in the body of the text, confirming that $P(\rho) = P^*$, $\mu_1(\rho) = \mu_2(\rho) = \mu^*$ and $c(\mu_a,\rho) = c^*(\mu_a)$ is a Markov equilibrium. □

This proposition suggests that differences in γ, rather than differences in effective discount factors, are what allow reputation to play a role: in the absence of the former, the static Markov equilibrium — a pooling equilibrium where both types have the same strategy — remains a Markov equilibrium in the dynamic game. (However, we note that this result does not rule out the existence of other equilibria where reputational considerations do matter.) In contrast, in Section 7, we show that if $\gamma_1 > \gamma_2$ (or one type is more hawkish than the other), as long as Assumption 1 holds, all Markov equilibria are separating. Thus reputation about hawkishness — the extent to which the central bank dislikes inflation — matters.

6. A reference game

To help distinguish between the effects of reputation directly and the effects of central banks attempting to manipulate their reputations, we introduce the following reference game. In it, the type of central bank is forever fixed, but is revealed immediately after households choose their price at date $t = 0$.

At date $t = 1$, after the central bank's type is revealed, this reference game is exactly the game analyzed in the Section 4 — the type of the central bank is known and unchanging. Let $(P^i, \mu^i, c^i(\mu_a))$ denote the equilibrium of this continuation game for a central bank of type i, where $g^i = \mathbb{E}\left[\frac{1}{c^i(\mu_a)}|\mu^i\right]$.[9]

For period $t = 0$, as a function of an initial reputation ρ_0, households choose a common price, $P_0(\rho_0)$. Given P_0 and the revealed type, the central bank of type i then chooses μ_0^i. Finally, after μ_a is realized, households facing a central bank of type i choose consumption $c_0^i(\mu_a)$.

Working backwards, and using condition (3), the household consumption decision $c_0^i(\mu_a)$ must satisfy[10]

$$c_0^i(\mu_a) = \frac{1+\mu_a}{P_0(\rho_0)}.$$

The optimal choice of μ_0^i satisfy

$$\mu_0^i \in \arg\max_{\mu\in[0,\overline{\mu}]} \int \left(\log(c_0^i(\mu_a)) - \alpha c_0^i(\mu_a)\right) dF(\mu_a|\mu) - \gamma_i h(\mu).$$

Using the equilibrium consumption, this is equivalent to

$$\mu_0^i \in \arg\max_{\mu\in[0,\overline{\mu}]} \mathbb{E}[\log(1+\mu_a)|\mu] - \alpha\frac{\mathbb{E}[1+\mu_a|\mu]}{P_0(\rho_0)} - \gamma_i h(\mu).$$

If $\partial\mathbb{E}[1+\mu_a|\mu]/\partial\mu \geq 0$, then it follows that the optimal choice μ_0^i is increasing in $P_0(\rho_0)$. Reputation considerations only affect the equilibrium choice of μ_0^i through the price level $P_0(\rho_0)$. That is, in this reference game, the central bank's type is public when it chooses μ_0^i, but it nevertheless reacts to the price level $P_0(\rho_0)$ which does depend on the initial probability the central bank is the hawkish type. If the initial reputation causes households to choose a higher (lower) price at date $t = 0$, both types of central banks will choose higher (lower) expected inflation in response.

This optimal pricing decision of households, $P_0(\rho_0)$, then boils down in this case to

$$P_0(\rho_0) = \alpha\frac{\sigma}{\sigma-1}\frac{1}{\beta}\left(\frac{\rho_0\mathbb{E}[1+\mu_a|\mu_0^1] + (1-\rho_0)\mathbb{E}[1+\mu_a|\mu_0^2]}{\rho_0\mathbb{E}\left[\frac{1}{1+\mu_a}|\mu^1\right] + (1-\rho_0)\mathbb{E}\left[\frac{1}{1+\mu_a}|\mu^2\right]}\right)$$

Suppose that $\mathbb{E}[1+\mu_a|\mu]$ is increasing in μ and that $\mathbb{E}[1/(1+\mu_a)|\mu]$ is decreasing in μ. Suppose also that in the equilibrium with certain types, $\mu^2 > \mu^1$. Consider now the effect of a change in reputation ρ_0 in the household's optimal price. Fixing μ_0^1 and μ_0^2, and assuming that $\mu_0^1 < \mu_0^2$, a low reputation causes households to expect relatively high inflation and thus set a high price in the initial

[9] P^i, μ^i, and $c^i(\mu_a)$ satisfy the same equations as for $P^\star, \mu^\star, c^\star(\mu_a)$ in the previous section, but for two distinct values of γ:

$$P^i = \alpha\frac{\sigma}{\sigma-1}\frac{1}{\beta}\frac{\mathbb{E}\left[1+\mu_a|\mu^i\right]}{\mathbb{E}\left[\frac{1}{1+\mu_a}|\mu^i\right]},\quad c^i(\mu_a) = \frac{1}{\alpha}\frac{\sigma-1}{\sigma}\beta\mathbb{E}\left[\frac{1}{1+\mu_a}|\mu^i\right]\frac{1+\mu_a}{\mathbb{E}[1+\mu_a|\mu^i]},\text{ and}$$

$$\mu^i \in \arg\max_{\mu\in[0,\overline{\mu}]} \int \left(\log(c^i(\mu_a)) - \alpha c^i(\mu_a)\right) dF(\mu_a|\mu) - \gamma_i h(\mu).$$

[10] Note that the household optimality condition is $c_0^i(\mu_a) = \frac{1+\mu_a}{P_0(\rho_0)}\min\left\{\frac{P^i}{\beta}\frac{1}{g^i}, 1\right\}$. However, we know that from period $t = 1$, the cash in advance constraint binds, and thus $\frac{P^i}{\beta}\frac{1}{g^i} \geq 1$. And this implies that the cash in advance also binds in period $t = 0$.

period. Similarly, a high reputation causes households to expect relatively low inflation and thus set a low price. This is absent any consideration by the central bank of how current inflation will affect its future reputation.

Thus this reference game lets us isolate the *direct* effect of reputation present here, as opposed to the effect of a central bank *worrying* about its reputation present in the full game.

7. Existence

In this section, we consider whether pure strategy symmetric Markov perfect equilibria exist. Our main point is that without sufficient noise disconnecting a central bank's action to the observed outcome, pure strategy symmetric Markov equilibria may **not** exist.

Our results regarding pooling equilibria are presented in the form of two propositions. In the first, we show there cannot be a pure strategy symmetric Markov equilibrium where the types pool at the stationary reputation, (or where $\mu_1(\rho^*) = \mu_2(\rho^*)$, where $\rho^* \equiv \frac{\epsilon}{\delta+\epsilon}$). This rules out, in particular, Markov strategies where the types *always* pool (or where $\mu_1(\rho) = \mu_2(\rho)$ for all ρ).

In the second proposition, we show that there cannot be a pure strategy symmetric Markov equilibrium where $\mu_1(\rho) = \mu_2(\rho)$ for *any* ρ, unless $\mu_1(\rho) = \mu_2(\rho) = 0$. This rules out equilibria where the types *ever* pool, as long as both $\mu_1(\rho)$ and $\mu_2(\rho)$ are positive for all ρ.

At the end, we argue that it is difficult to sustain a separating pure strategy equilibria without sufficient noise disconnecting a central bank's target inflation from realized inflation, (and with sufficient patience).

First, we show there exists no Markov equilibria in pure strategies where the types pool at $\rho = \rho^*$.

Proposition 2. *Suppose a pure symmetric Markov strategy* $(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho))$ *has* $\mu_1(\rho) = \mu_2(\rho) = \mu$ *for* $\rho = \rho^* \equiv \frac{\epsilon}{\delta+\epsilon}$. *If Assumption 1 holds,* $(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho))$ *is not a symmetric Markov equilibrium.*

Proof. Towards a contradiction, consider that there is such an equilibrium. Note that $\rho^+(\mu_a, \rho^*) = \rho^*$ for all μ_a, since $\hat{\rho}(\mu_a, \rho) = \rho^*$ from $\mu_1(\rho^*) = \mu_2(\rho^*) = \mu$, and that $\rho^+(\rho^*, \mu_a) = (1-\delta)\rho^* + \epsilon(1-\rho^*) = \rho^*$. From (3), then,

$$\frac{P(\rho^*)c(\mu_a, \rho^*)}{1+\mu_a} = \min\left\{\frac{P(\rho^*)}{\beta g(\rho^*)}, 1\right\}.$$

This implies that starting from $\rho = \rho^*$, the cash-in-advance constraint always binds or never does.

Using the same argument as before, we can rule out that it never does, and it follows that

$$\frac{P(\rho^*)c(\mu_a, \rho^*)}{1+\mu_a} = 1.$$

From the definition of $g(\rho)$, we have

$$g(\rho^*) = \mathbb{E}\left[\frac{1}{c(\mu_a, \rho^*)}|\mu\right].$$

Using (2), we get that

$$P(\rho^*) = \frac{\alpha}{\beta}\frac{\sigma}{\sigma-1}\frac{\mathbb{E}[1+\mu_a|\mu]}{\mathbb{E}[1/(1+\mu_a)|\mu]}.$$

Now, given that $\rho^+(\rho^*)$ is independent of μ_a, the first order condition for type i is

$$\frac{\partial \mathbb{E}[\log(1+\mu_a)|\mu]}{\partial \mu} - \frac{\sigma-1}{\sigma}\beta\frac{\mathbb{E}[\frac{1}{1+\mu_a}|\mu]}{\mathbb{E}[1+\mu_a|\mu]}\frac{\partial \mathbb{E}[1+\mu_a|\mu]}{\partial \mu} \leq \gamma_i h_i'(\mu)$$

with equality if $\mu > 0$. This equation cannot be satisfied with $\mu = 0$, as it violates Assumption 1. Nor can it hold for $\mu > 0$, as $\gamma_1 \neq \gamma_2$. □

This proposition rules out that in an equilibrium, the types pool on any subset of ρ that contains ρ^*. We can strengthen this result, but focusing on *interior* equilibria.

Proposition 3. *Suppose a Markov pure strategy* $(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho))$ *has* $\mu_1(\rho) = \mu_2(\rho) > 0$ *for some* $\rho \in [\epsilon, 1-\delta]$. *Then,* $(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho))$ *is not a symmetric Markov equilibrium.*

Proof. Suppose $(P(\rho), \mu_1(\rho), \mu_2(\rho), c(\mu_a, \rho))$ has $\mu_1(\rho) = \mu_2(\rho) > 0$ for some $\rho \in [\epsilon, 1-\delta]$. Then, $\hat{\rho}(\rho, \mu_a, \mu_1(\rho), \mu_2(\rho)) = \rho$ and $\rho^+(\rho, \mu_a, \mu_1(\rho), \mu_2(\rho)) = (1-\delta)\rho + \epsilon(1-\rho)$. That is, each is independent of the realization of μ_a. Condition (5) can then be written as

$$\frac{\partial \mathbb{E}[\log(c(\mu_a, \rho)) - \alpha c(\mu_a, \rho)|\mu = \mu_i(\rho)]}{\partial \mu} = \gamma_i h'(\mu_i). \tag{15}$$

Since $\gamma_2 < \gamma_1$ and $h'(\mu) > 0$ for $\mu > 0$, Eq. (15) cannot simultaneously hold for both $i = 1$ and $i = 2$; thus, a contradiction is generated. □

Next, we argue (informally) that noise makes the existence of a separating equilibrium, for given values of β and ϵ, easier.

To see this, note that if $\mu_1(\rho) \neq \mu_2(\rho)$ and there is no noise, or $\mu_a = \mu_i(\rho)$ with certainty, Bayesian updating requires that $\rho^+(\mu_1(\rho), \rho) = 1 - \delta$ (the highest possible reputation) and $\rho^+(\mu_2(\rho), \rho) = \epsilon$ (the lowest possible reputation). This implies that in order for the dovish central bank not to wish to mimic the hawkish central bank,

$$\begin{aligned}&(1-\beta_2)[\log(c(\mu_2(\rho), \rho)) - \alpha c(\mu_2(\rho), \rho)] + \beta_2 V_2(\epsilon)\\ &\quad \geq (1-\beta_2)[\log(c(\mu_1(\rho), \rho)) - \alpha c(\mu_1(\rho), \rho)] + \beta_2 V_2(1-\delta).\end{aligned}$$

Rearranged, this becomes

$$\begin{aligned}&(1-\beta_2)[\log(c(\mu_2(\rho), \rho)) - \alpha c(\mu_2(\rho), \rho) - (\log(c(\mu_1(\rho), \rho)) - \alpha c(\mu_1(\rho), \rho))]\\ &\quad \geq \beta_2(V_2(1-\delta) - V_2(\epsilon)).\end{aligned}$$

In words, the left hand side is the static gain to the dovish central bank type of following its strategy $\mu_2(\rho)$ as opposed to mimicking the hawkish central bank, while the right hand side is the cost, in terms of the forgone value of attaining the reputation of being a hawkish central bank, of following strategy $\mu_2(\rho)$. As long as $V_2(1-\delta) > V_2(\epsilon)$, for high values of β_2 (either because β is high or because ϵ is low), this incentive condition becomes difficult to satisfy.

So instead assume noise, so μ_a depends on but is not necessarily equal to $\mu_i(\rho)$. Then, the equivalent condition to 3 is

$$\begin{aligned}&(1-\beta_2)(\mathbb{E}[\log(c(\mu_a, \rho)) - \alpha c(\mu_a, \rho)|\mu_2(\rho)] - \mathbb{E}[\log(c(\mu_a, \rho)) - \alpha c(\mu_a, \rho)|\mu_1(\rho)])\\ &\quad \geq \beta_2(\mathbb{E}[V_2(\rho^+(\mu_a, \rho)|\mu_1(\rho))] - \mathbb{E}[V_2(\rho^+(\mu_a, \rho)|\mu_2(\rho))]).\end{aligned}$$

Now, the effect on type $i = 2$'s reputation of mimicking type $i = 1$ is not so extreme. In fact, if μ_a were determined only by noise, tomorrow's reputation, ρ^+, would be independent of μ_a, the right hand side of 3 would be zero, and the condition would be satisfied merely by making type $i = 2$ have a static incentive to follow $\mu_2(\rho)$ rather than $\mu_1(\rho)$.

8. An example equilibrium

In the previous section, we established that without sufficient noise, it is difficult for pure strategy Markov equilibria to exist. In this section, we present an example where the central bank's target growth rate, μ_i, is sufficiently disconnected from the actual money growth rate that a pure strategy symmetric Markov equilibrium exists.

In the example, we let $\alpha = 1$ (so the efficient level of consumption $c^* = 1$), $\sigma = 5$, and $\beta = 0.99$. For the central banks, we let $h(\mu_i) = 0.5\mu_i^2$ and let the switching probabilities be $\epsilon = \delta = 0.02$.

The determination of actual money growth, μ_a, is as follows: with probability μ_i, μ_a is determined by an exponential distribution with mean 0.04. With complementary probability $1 - \mu_i$, μ_a is determined by an exponential distribution with mean 0.01. Given these, the inflation cost parameters, γ_1 and γ_2, are chosen so that if there were only one type, in equilibrium, the hawkish type, type $i = 1$, would set μ_1 so that mean inflation was 2% each period, while the dovish type, type $i = 2$, would set μ_2 so that mean inflation was 3% each period. That is, households are trying to determine if the central bank has a mean inflation target of 2% or 3%.[11]

Under these parameters, in a world where it is common knowledge that central banks are always the hawkish ($i = 1$) type, in every period

$$P = 1.313, \quad \mathbb{E}[\log(c) - \alpha c] = -1.0297.$$

In a world where it is common knowledge that central banks are always the dovish ($i = 2$) type, in every period

$$P = 1.338, \quad \mathbb{E}[\log(c) - \alpha c] = -1.0320.$$

For comparison, in the efficient allocation where $c = \frac{1}{\alpha} = 1$ every period, $\log(c) - \alpha c = -1$. The equilibrium price level set by households, Eq. (8), is higher when it is common knowledge the central bank is dovish since expected money growth, and thus inflation, is higher when the central bank is dovish, and households cannot spend the money received in the current period until the following period.

Prices as a function of reputation are shown in Fig. 1. In this figure, the lower curve represents prices set by sellers in our actual game. Here, households are always unsure of the type of the central bank. Central banks know this, and thus strategically set $\mu_1(\rho)$ and $\mu_2(\rho)$ to influence their future reputation. For reference, the top dashed line represents the price charged by sellers if they believe the central bank is the hawkish type with probability ρ, but where this type is revealed just after the price is set and will never change again, our reference game from Section 5. Under this assumption, a central bank of type i will always choose its static best response given the prices set. We include this to see the effect of reputation *directly*, as opposed to the effects of central banks *caring* about their reputation. For all figures, dashed lines represent the outcome in the reference game, while solid lines represent the outcome for the actual game. Note also that in each figure, reputations go from the lowest possible beginning-of-period reputation, ϵ, to the highest possible beginning-of-period reputation, $(1 - \delta)$.

Fig. 2 displays $\mu_1(\rho)$ and $\mu_2(\rho)$ as a function of reputation, again for the full game and the reference game, where types are revealed once and for all just after prices are set. Here, type 2 takes a higher action $\mu_2(\rho)$ than type 1 for all ρ, but both types i

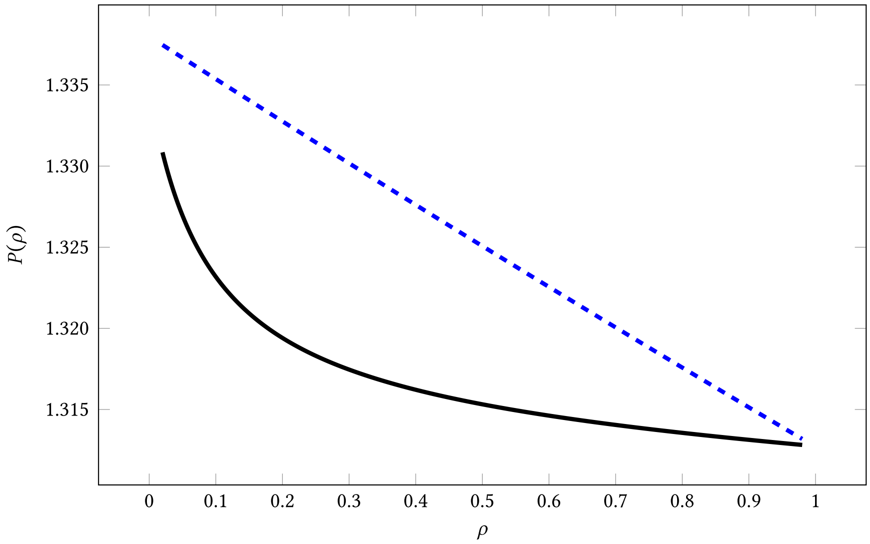


Fig. 1. Price as a function of reputation.
Note: The solid line is the equilibrium in the actual game. The dashed line is the outcome in the reference game.

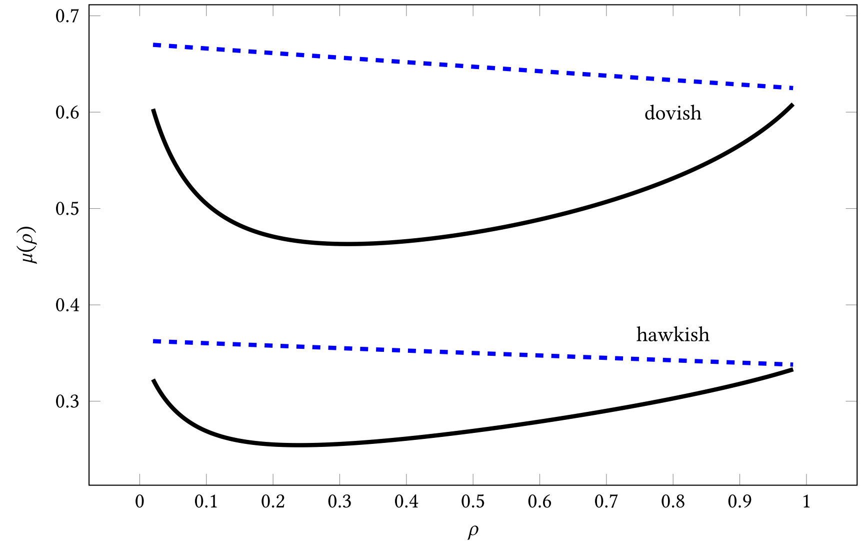
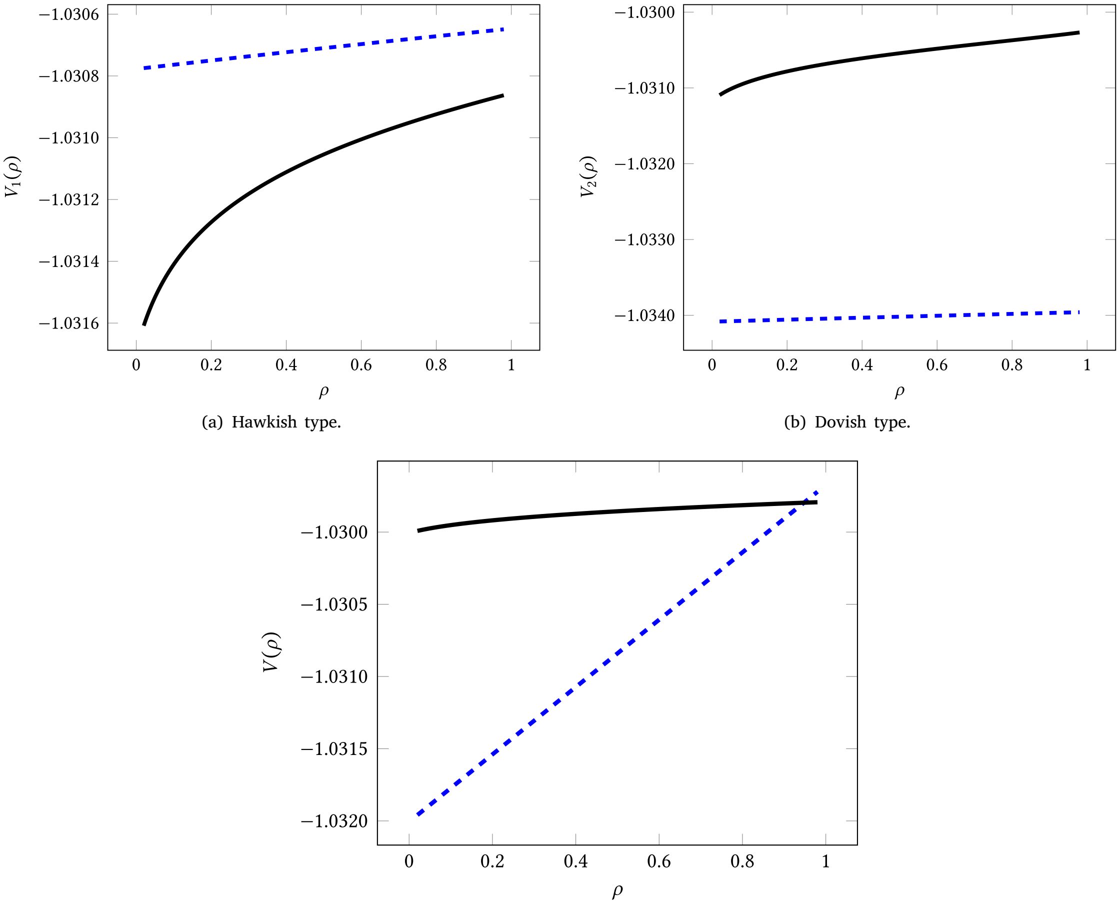


Fig. 2. Central bank actions as functions of reputation.
Note: The solid lines are from the equilibrium in the actual game. The dashed lines are from the reference game.

set μ_i less than they would if they were unconcerned with protecting their reputation. Notably, in this example, both types choose their lowest action values when reputation is interior, when Bayesian posteriors are most easily moved.

Fig. 3 displays the value functions for both the hawkish and dovish central bank types in panels (a) and (b) respectively. Both types value being seen as more likely to be the hawkish type. Interestingly, the hawkish type is made worse off from the continuing uncertainty of households regarding central bank type, while the dovish type is made better off.

[11] For the numerical analysis, we approximate the exponential distributions with a discrete distribution in a grid. All computations are in the Github repository: https://github.com/manuelamador/monetary_reputation/.



(a) Hawkish type.

(b) Dovish type.

(c) Households.

Fig. 3. Payoffs as functions of reputation.
Note: The solid lines are from the equilibrium in the actual game. The dashed lines are from the reference game.

In Fig. 3(c), we present the expected discounted utility of households as a function of beginning of period reputation. Not surprisingly, a household's expected discounted utility is increasing in the probability of its facing a hawkish central bank type. Furthermore, unless reputation is near its highest possible value, households are better off in the world with continuing uncertainty (solid line) versus the world where all uncertainty is revealed after the initial price is set, again because continued uncertainty offsets the inflation temptations of both central bank types.

Another question is whether reputation considerations persist. That is, nothing theoretically rules out the possibility that the central bank's reputation, ρ, is nearly always near its extremes, ϵ and $1-\delta$. Figs. 4 and 5 demonstrate that at least for these parameters, our model exhibits a long lasting effect of reputation: reputations do not quickly go to either pole. Fig. 4 gives the expected paths of reputation, starting from $\rho = 0.5$ conditional on being the hawkish type (top path), dovish type (bottom) path, and unconditionally (middle path). Note first that even conditional on being a hawkish or dovish type, expected reputation remains above $\rho = 0.38$ (for the dovish type) and below $\rho = 0.63$ (for the hawkish type). Noise and the fact that types possibly switch keeps expected reputations interior. Fig. 5 displays the ergodic distribution of reputation, ρ, again displaying that interior reputations persist.

Fig. 6 plots the ergodic distribution of inflation and further displays the effect of the two central bank types caring about their reputation. In the equilibrium where it is common knowledge the central bank is hawkish, inflation is on average 2%, and in the equilibrium where it is common knowledge the central bank is dovish, inflation is on average 3% (and, since, $\epsilon = \delta$, each type is present half the time). But when types are hidden, and thus reputation is coveted, inflation in our example is on average 2.1%.

Back to the central question posed in the introduction — how "should" a hawkish central bank act to recover a hawkish reputation if it has lost one, or maintain a hawkish reputation if it has one. The answer is in its Markov best response function in Fig. 2. First, not surprisingly, a hawkish type should inflate less than it would if it were the dovish type (or the lower solid line is lower than

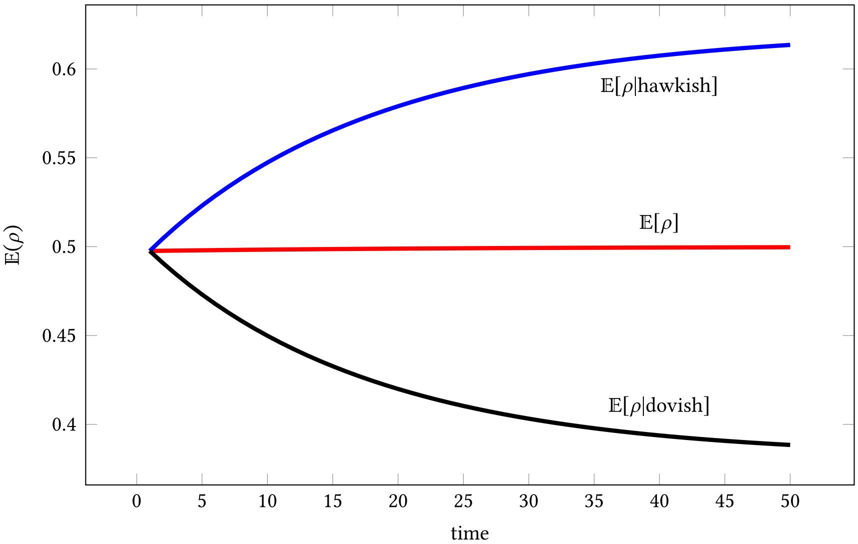


Fig. 4. Expectation of ρ over time starting from $\rho = .5$.

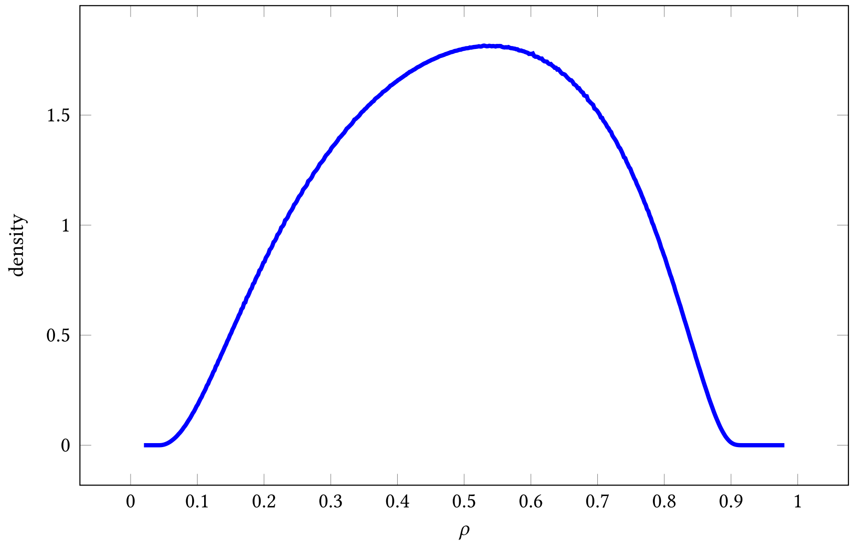


Fig. 5. Ergodic distribution of reputation, ρ.

the higher solid line). Second, again not surprisingly, it should inflate less than it would absent worrying about its reputation (or the lower solid line is lower than the lower dashed line). Third, and perhaps surprisingly, its Markov best response is U shaped in its reputation. If it has a very high or very low reputation, it should inflate more than if it has a middling reputation. This comes directly from Bayes' rule. For very high or low reputations, the effect on the bank's future reputation of any given money growth realization is lower — it's hard to move an extreme belief — but the benefits to inflating, holding constant the current price chosen by households, remain the same. This implies that a hawkish central bank with a high reputation can "afford" (again in a Markov best response sense) to inflate more than one with a more moderate reputation. The hawkish central bank with the middling reputation should be more hawkish. A hawkish central bank with a very low reputation (say because of a long series of high actual inflation outcomes) should also act not relatively hawkish, here because it doesn't do that much good until nature delivers enough lower inflation outcomes to deliver a more moderate reputation (which will happen eventually on average).

To assess robustness, we varied the main parameters of the model: the common discount rate β, the elasticity of substitution parameter σ, the penalties for each type, γ_1 and γ_2, and the switching probabilities, ϵ and δ, from their benchmark values. In each case, the general patterns observed in Figs. 1 through 5 remain the same. Furthermore, the effects of varying each parameter in isolation are generally as expected: If the discount factor β or the elasticity of substitution σ decreases, each central bank inflates more for any given reputation, and prices are correspondingly higher. If the penalty γ_1 for the hawkish type decreases (or the

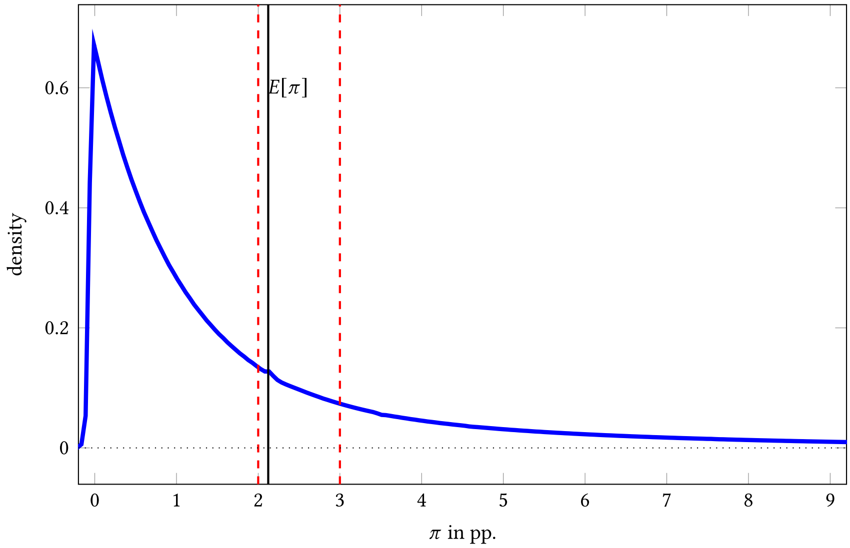


Fig. 6. Ergodic distribution of inflation.
Note: Inflation rates are computed using (6) over the ergodic distribution of ρ. The solid vertical line marks the average inflation under this distribution. The dashed vertical lines at 2% and 3% indicate the mean inflation levels that the hawkish and dovish types, respectively, would select if each were the only type.

hawkish type becomes less hawkish) both central bank types inflate more, and prices are higher for any given reputation. If the penalty γ_2 for the dovish type decreases (or the dovish type becomes more dovish), the dovish type inflates more while, somewhat unexpectedly, the hawkish type inflates less, leaving prices little changed, but overall higher for each reputation. Finally, if the probability that a hawkish type is replaced by a dovish type, δ, or the probability that a dovish type is replaced by a hawkish type, ϵ, increases, prices rise and both types inflate more for any given reputation. In both cases, the increase in switching probabilities acts somewhat similarly to a decrease in the discount parameter, β. If the survival probability of the hawkish type decreases, this has a direct effect of the hawkish type inflating more, and an indirect effect of the dovish type inflating more because the hawkish type is. Likewise, if the survival probability of the dovish type decreases, this has a direct effect of the dovish type inflating more, and an indirect effect of the hawkish type inflating more because the dovish type is.

While performing these exercises, we never found a case where the dovish type inflates less than the hawkish type, even if the hawkish type is extremely impatient (that is, a high δ) and the dovish type is patient (that is, a low ϵ). We think the reason for this numerical result may be contained in Proposition 1 which suggests that reputation building in this model is about the inflation costs and not about the effective discount factors. In the game where the inflation costs are different, impatience rates matter. But this has a limit as when inflation costs are the same, the central banks impatience rates become irrelevant.

9. Conclusion

This paper presents a model of central banks whose actions noisily affect their reputation. In it, we argue that if there is sufficiently little discounting, a pure strategy Markov perfect equilibrium cannot exist if there isn't enough noise. However, we show computationally that if there is enough noise, a pure strategy Markov perfect equilibrium can exist with appealing characteristics: both hawkish and dovish central bank types inflate less than they otherwise would, especially when their reputation is middling, and this outcome improves household welfare.

Data availability

No data was used for the research described in the article.

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